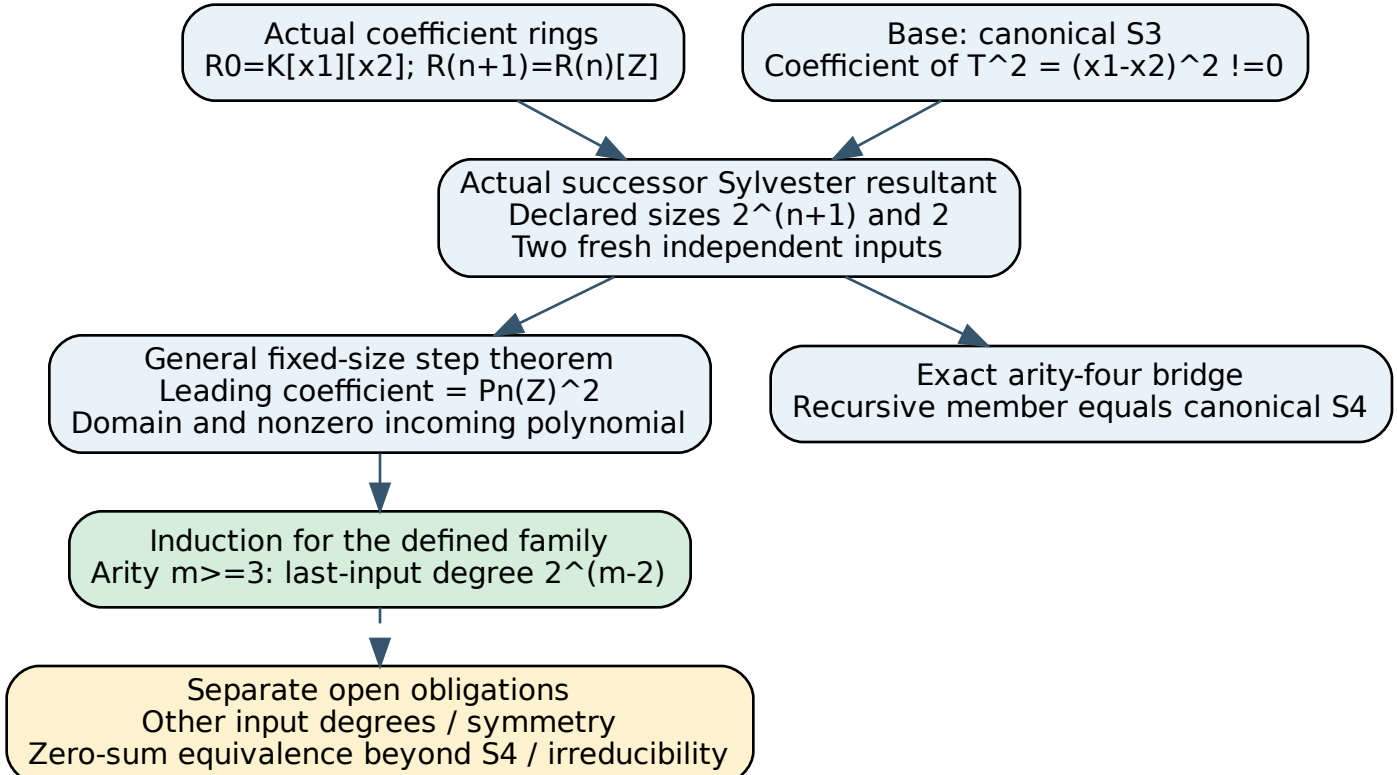
Actual coefficient rings
R0=K[x1][x2]; R(n+1)=R(n)[Z]
Base: canonical S3
Coefficient of T^2 = (x1-x2)^2 !=0
Actual successor Sylvester resultant
Declared sizes 2^(n+1) and 2
Two fresh independent inputs
General fixed-size step theorem
Leading coefficient = Pn(Z)^2
Domain and nonzero incoming polynomial
Exact arity-four bridge
Recursive member equals canonical S4
Induction for the defined family
Arity m>=3: last-input degree 2^(m-2)
Separate open obligations
Other input degrees / symmetry
Zero-sum equivalence beyond S4 / irreducibility